

PILLI SANTHOSH REDDY

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Profile

Mechanical Engineering graduate with a strong interest in motorsport engineering, backed by a solid foundation in material science and structural analysis. Proficient in industry-relevant tools including CAD and FEA software for performance-driven design and simulation. Experienced in the complete engineering lifecycle, from concept development to prototyping and testing, with a focus on optimizing performance, reliability, and efficiency. Seeking an entry-level opportunity in motorsport or automotive engineering to apply analytical skills and contribute to high-performance vehicle development.

Education

09/2022 – 04/2026	B.Tech
Nuzvid	<i>Rajiv Gandhi University of Knowledge and Technologies</i>
	Mechanical Engineering
	CGPA : 7.5
01/2021 – 09/2022	Pre-University course
Nuzvid	<i>Rajiv Gandhi University of Knowledge and Technologies</i>
	MPC
	CGPA : 8.3

Skills

Core Skills

CAD: SolidWorks, Catia
CAE: ANSYS Workbench

Motorsport & Performance Engineering

Vehicle Dynamics Fundamentals
Sim Racing & Telemetry Analysis
Python & Machine learning

Soft Skills

Strong problem solving and analytical skills.
Communication Skills
Team & Time Management

Projects

Sim Racing Telemetry Analysis

Collected and analyzed telemetry data from racing simulators such as **Assetto Corsa Competizione**. Evaluated braking points, throttle application, steering inputs, and sector times to identify performance improvements. Compared multiple laps using telemetry software **MoTec I2** to optimize driving consistency and lap time performance. Applied data-driven techniques similar to professional motorsport performance engineering workflows.

SAE Baja Chassis Crash Test & Safety Analysis

Performed non-linear dynamic **FEA crash simulations** using **ANSYS Mechanical** to validate driver safety and meet rigorous SAE impact standards. Analyzed high-rate impact scenarios to assess localized stress concentration, energy absorption, and structural deformation in the roll cage.

Lap Time Prediction Using Machine Learning (Python)

Built a predictive model using telemetry parameters such as speed, throttle position, braking, and sector times to estimate vehicle lap times. Applied data preprocessing, feature selection, and regression techniques in Python using pandas and scikit-learn. Evaluated model performance using standard regression metrics and visualized results to identify performance trends.

Volunteer experience

- 08/2024 – 04/2026 **Automobile Club Co-ordinator**
Student Co-ordinator
As the Co-ordinator of the Automobile Club, I directed all strategic operations, managed a diverse team of officers, and oversaw the planning and execution of all workshops, competitions, and events. This role helped me to develop strong **leadership**, **financial oversight** for club projects, and effective **liaison** with faculty and external sponsors.
- 10/2024 – 10/2025 **Bureau of Indian Standards**
Student President
As BIS Club President, I directed the club's strategy and enhanced member skills by personally conducting numerous workshops and educational classes to my fellow students
- 11/2023 – 08/2025 **Helping Hands**
Branch Representative
As a **Helping Hands Representative**, I coordinated and executed community service projects, mobilizing volunteers for local initiatives. I served as the primary liaison between our organization and partner charities, ensuring our efforts created meaningful, positive change.

Certificates

Applied race car vehicle dynamics
from Indian Motorsports Academy

AutoCAD certification
from intrainz

GD&T certification
from udemy